COMP90050: Advanced Database Systems

THE UNIVERSITY OF MELBOURNE

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Database architectures

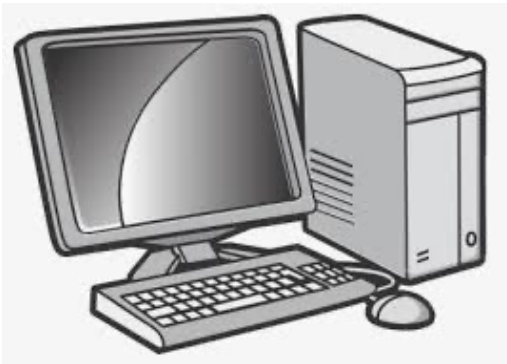
Week 1

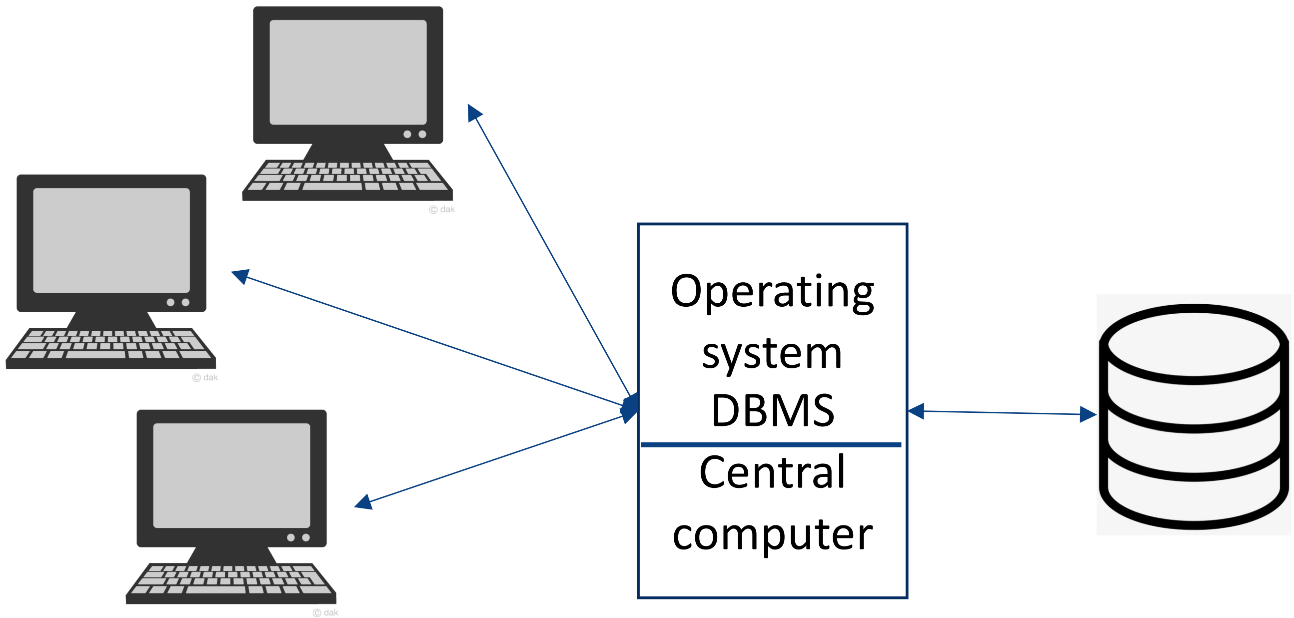
Database Architectures

Centralized	Distributed	WWW	Grid	P2P	Cloud
• Data stored in one location	• Data distributed across several nodes, can be in different locations	• Stored all over the world, several owners of the data	• Like distribu-ted, but each node manages own resource; system doesn't act as a single unit.	• Like grid, but nodes can join and leave network at will (unlike Grid)	• Generalization of grid, but resources are accessed on-demand

Database Architectures

Centralised database systems

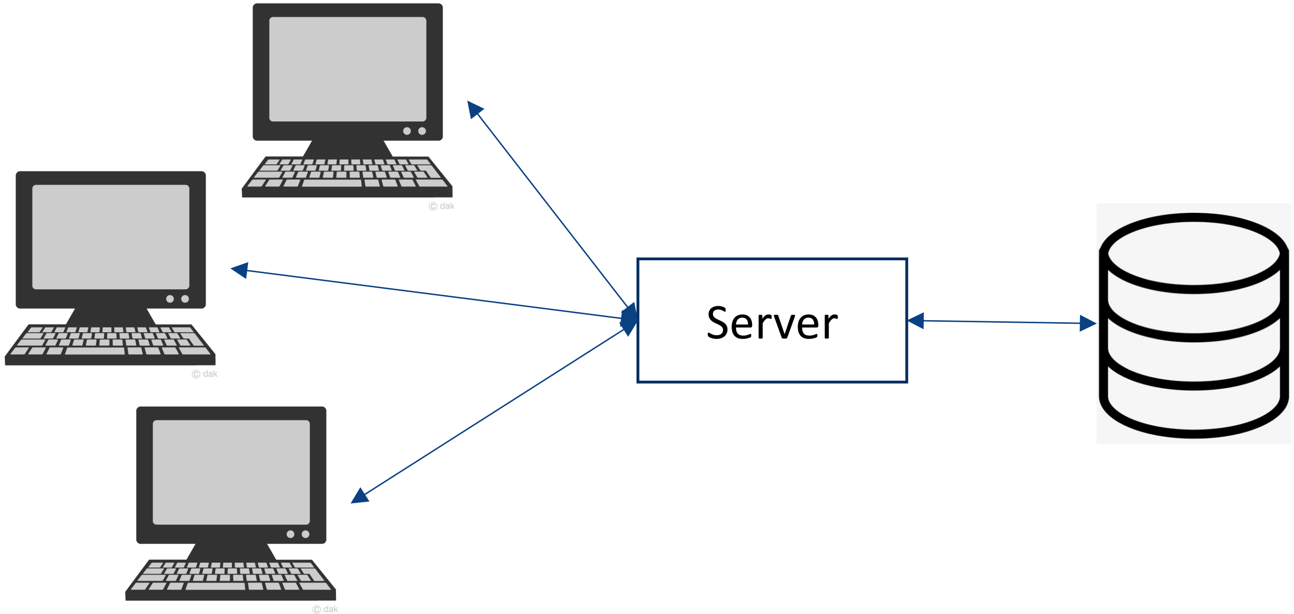




- Data in one location
- System administration is simple
- Optimisation process is generally very effective
- PC/Cluster Computing/data centres are examples of this

Database Architectures

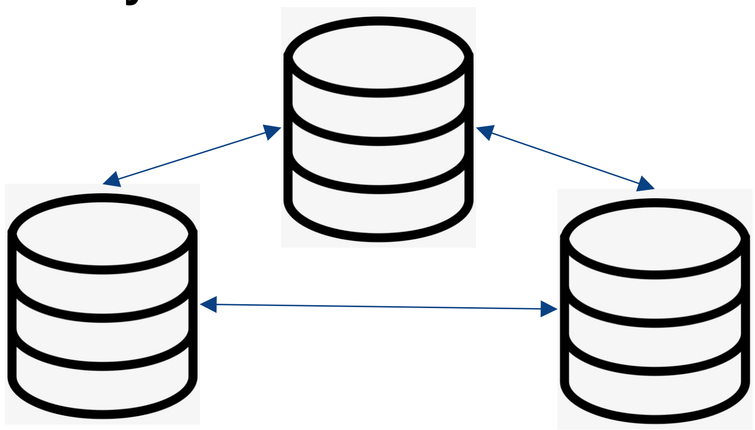
Client-server architecture that uses a centralized database server-



- A central server with a central DB in one location
  (but client and server can be in different locations)
- Client generally provides user interfaces for input and output
- Server provides all the necessary database functionality
- System administration is relatively simple
- System recovery is similar to centralised systems

Database Architectures

Distributed database systems:



- Data is distributed across several nodes in different locations
- Nodes are connected by network
- System provides concurrency, recovery and transaction processing
- System administration is very hard – usually with a single resource manager
- Crash recovery is complicated
- There are usually data replication -- potential inconsistency

Database Architectures

World Wide Web

- Data is stored in many locations
- Several owners of data - no certainty of data availability or consistency
- Optimisation process is generally very ineffective
- Evolving database technology -- no standards have been developed except in case of XML/http and some protocols for accessing data
- Security could be a potential problem
- Notions of Transactions is much more difficult to enforce

Database Architectures

Grid Computing and Databases

- Very similar to distributed database systems
- Data and processing are shared among a group of computer systems which may be geographically separated.
- Usually designed for particular purpose – e.g., a scientific application
- **Administration of such systems are done locally by each owner of the system**
- Reliability and security of such systems are not well developed or studied.
- Grid systems model is more or less outdated by Cloud Computing model

Database Architectures

Cloud computing

https://youtu.be/mxT233EdY5c

Database Architectures

Cloud computing

Cloud computing offers online computing, storage and a range of new services for data and devices that are accessible through the Internet.

User pays for the services just like phone services, electricity, etc.

Huge potential for developing applications with minimal infrastructure costs

Cloud services offered in several forms:

- Iaas – Infrastructure as a service (provide virtual machines)
- Paas – Platform as a service (Provide environment like Linux, Windows, etc.)
- Saas – Software as a service (Specific applications like RDB, Mail, etc.)

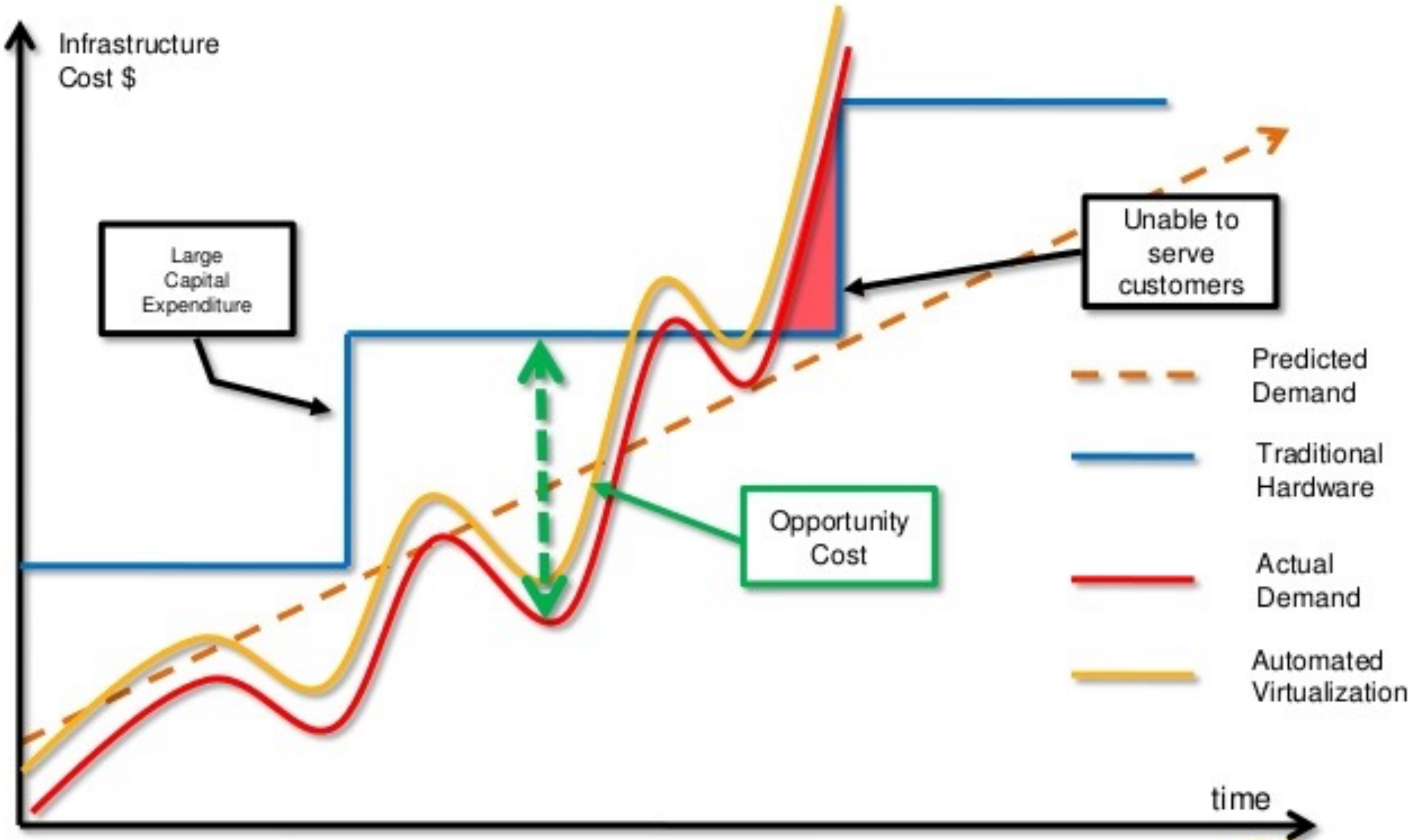
An example: AWS: Amazon Web Services

Attributes of Cloud Computing

- No capital expenditure
- Pay as you go and pay only for what you use
- True elastic capacity; Scale up and down
- Improves time to market
- You get to focus your engineering resources on what differentiates you vs. managing the undifferentiated infrastructure resources

amazon webservices™

Elastic and Pay-Per-Use Infrastructure



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Database Architectures

P2P Databases

- Data and processing is shared among a group of computer systems which may be geographically separated as in Grid Database systems
- **Computer nodes can join and leave the network at will unlike in Grid Databases => much harder to design transaction models**
- Usually designed for particular usage – e.g. a scientific application
- Administration of such system is done by the owners of the data

Database Architectures

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Core Concepts of Database management system

Database performance metrics

- Efficiency/speed
- Effectiveness
- Security & Reliability

Efficiency

- Hardware
  - Disks and I/O bandwidth
  - Main memory
  - Type of architecture
- Software/ DB tuning
  - Types of DB
  - Indexing
  - Query optimisation

Effectiveness

- Concurrent users
- Transactions

Reliability

- Crash recovery
- Fault tolerance
- Data duplication